CV 5101 – Modelling, Uncertainty, and Data for Engineers

Assignment 1: Numerical Modelling

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1 Evaluating Finite Difference Approximations at Specific Nodes

Let us analyze the following basic trigonometric function:

$$f(x) = \sin x$$

The exact, analytical derivative of this function is known to be:

$$f'(x) = \cos x$$

In this exercise, we will approximate the derivative at three distinct points: $x = \frac{\pi}{4}$, $x = \frac{\pi}{2}$, and $x = \pi$. This evaluation will be performed utilizing the forward, backward, and central finite difference schemes.

For our manual calculations, we maintain a uniform step size of $h = 0.1$ across all three approaches. The mathematical formulations for these finite difference methods are defined as follows:

Forward Difference Scheme

$$f'_F(x) = \frac{f(x+h) - f(x)}{h}$$

Backward Difference Scheme

$$f'_B(x) = \frac{f(x) - f(x-h)}{h}$$

Central Difference Scheme

$$f'_C(x) = \frac{f(x+h) - f(x-h)}{2h}$$

To quantify the accuracy of our approximations, the absolute error is computed using the formula:

$$E = \left| f'_{\text{numerical}}(x) - f'_{\text{exact}}(x) \right|$$

1.1 Calculation of Derivative at $x = \pi/4$

The exact derivative evaluated at this point is:

$$f'\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \approx 0.70710678$$

Forward Difference:

$$f'_F\left(\frac{\pi}{4}\right) = \frac{\sin(\frac{\pi}{4} + 0.1) - \sin(\frac{\pi}{4})}{0.1}$$

$$f'_F\left(\frac{\pi}{4}\right) = \frac{\sin(0.785398 + 0.1) - \sin(0.785398)}{0.1}$$

$$f'_F\left(\frac{\pi}{4}\right) = \frac{\sin(0.885398) - 0.70710678}{0.1}$$

$$f'_F\left(\frac{\pi}{4}\right) = \frac{0.77416708 - 0.70710678}{0.1} = \frac{0.06706030}{0.1} = 0.6706030$$

Consequently, the absolute error is:

$$E_F = |0.6706030 - 0.70710678| = 0.0365038$$

Backward Difference:

$$f'_B\left(\frac{\pi}{4}\right) = \frac{\sin(\frac{\pi}{4}) - \sin(\frac{\pi}{4} - 0.1)}{0.1}$$

$$f'_B\left(\frac{\pi}{4}\right) = \frac{0.70710678 - \sin(0.785398 - 0.1)}{0.1}$$

$$f'_B\left(\frac{\pi}{4}\right) = \frac{0.70710678 - \sin(0.685398)}{0.1}$$

$$f'_B\left(\frac{\pi}{4}\right) = \frac{0.70710678 - 0.63298131}{0.1} = \frac{0.07412547}{0.1} = 0.7412547$$

The corresponding absolute error is:

$$E_B = |0.7412547 - 0.70710678| = 0.0341479$$

Central Difference:

$$f'_C\left(\frac{\pi}{4}\right) = \frac{\sin(\frac{\pi}{4} + 0.1) - \sin(\frac{\pi}{4} - 0.1)}{2(0.1)}$$

$$f'_C\left(\frac{\pi}{4}\right) = \frac{0.77416708 - 0.63298131}{0.2}$$

$$f'_C\left(\frac{\pi}{4}\right) = \frac{0.14118577}{0.2} = 0.70592886$$

The absolute error yields:

$$E_C = |0.70592886 - 0.70710678| = 0.00117792$$

As demonstrated, the central difference method significantly minimizes the error at $x = \pi/4$.

1.2 Calculation of Derivative at $x = \pi/2$

The true analytical derivative is:

$$f'\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$$

Forward Difference:

$$f'_F\left(\frac{\pi}{2}\right) = \frac{\sin(\frac{\pi}{2} + 0.1) - \sin(\frac{\pi}{2})}{0.1}$$

$$f'_F\left(\frac{\pi}{2}\right) = \frac{0.99500417 - 1}{0.1} = \frac{-0.00499583}{0.1} = -0.04995835$$

Thus, the error is:

$$E_F = | - 0.04995835 - 0| = 0.04995835$$

Backward Difference:

$$f'_B\left(\frac{\pi}{2}\right) = \frac{\sin(\frac{\pi}{2}) - \sin(\frac{\pi}{2} - 0.1)}{0.1}$$

$$f'_B\left(\frac{\pi}{2}\right) = \frac{1 - 0.99500417}{0.1} = \frac{0.00499583}{0.1} = 0.04995835$$

Thus, the error is:

$$E_B = |0.04995835 - 0| = 0.04995835$$

Central Difference:

$$f'_C\left(\frac{\pi}{2}\right) = \frac{\sin(\frac{\pi}{2} + 0.1) - \sin(\frac{\pi}{2} - 0.1)}{0.2}$$

$$f'_C\left(\frac{\pi}{2}\right) = \frac{0.99500417 - 0.99500417}{0.2} = \frac{0}{0.2} = 0$$

Thus, the error is $E_C = 0$. Due to the symmetry of the sine function around its peak and our uniform step size, the central difference provides the precise exact derivative here.

1.3 Calculation of Derivative at $x = \pi$

The analytical derivative is:

$$f'(\pi) = \cos(\pi) = -1$$

Forward Difference:

$$f'_F(\pi) = \frac{\sin(\pi + 0.1) - \sin(\pi)}{0.1}$$

$$f'_F(\pi) = \frac{-0.09983342 - 0}{0.1} = -0.99833417$$

Error:

$$E_F = |-0.99833417 - (-1)| = |-0.99833417 + 1| = 0.00166583$$

Backward Difference:

$$f'_B(\pi) = \frac{\sin(\pi) - \sin(\pi - 0.1)}{0.1}$$

$$f'_B(\pi) = \frac{0 - 0.09983342}{0.1} = -0.99833417$$

Error:

$$E_B = |-0.99833417 - (-1)| = 0.00166583$$

Central Difference:

$$f'_C(\pi) = \frac{\sin(\pi + 0.1) - \sin(\pi - 0.1)}{2(0.1)}$$

$$f'_C(\pi) = \frac{-0.09983342 - 0.09983342}{0.2} = \frac{-0.19966684}{0.2} = -0.99833417$$

Error:

$$E_C = |-0.99833417 - (-1)| = 0.00166583$$

1.4 Summary of Numerical Approximations

The approximations and their respective errors for $h = 0.1$ are compiled in Table 1.

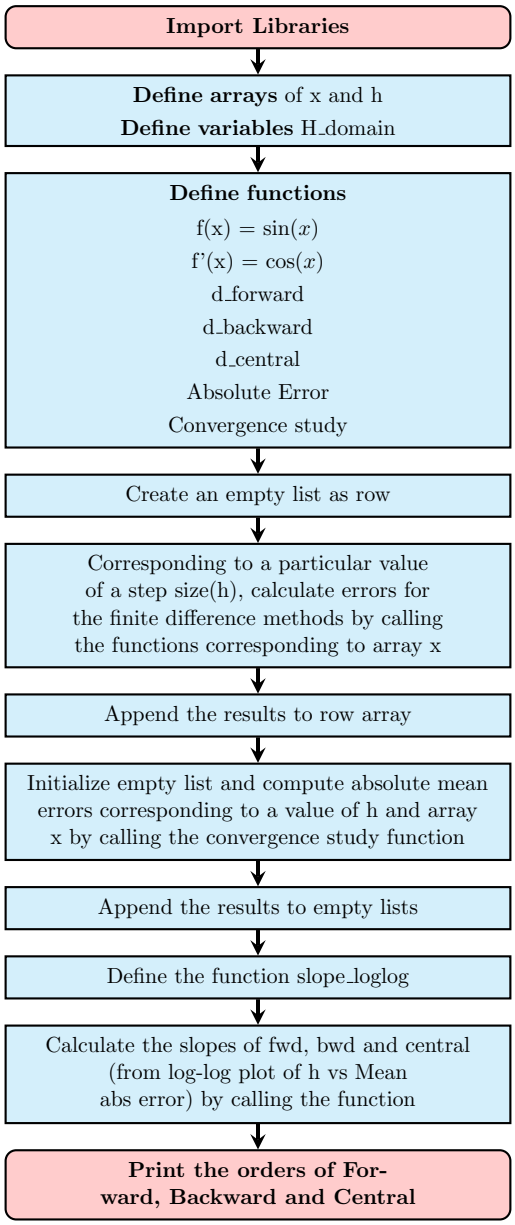
A clear trend emerges from the tabulated data: the central difference methodology consistently yields substantially lower errors compared to its forward and backward counterparts.

Table 1: Comparison of different finite difference approximations for $h = 0.1$.

x	Exact	Forward	E_F	Backward	E_B	Central	E_C
$\pi/4$	0.707107	0.670603	0.036504	0.741255	0.034148	0.705929	0.001178
$\pi/2$	0	-0.049958	0.049958	0.049958	0.049958	0	0
π	-1	-0.998334	0.001666	-0.998334	0.001666	-0.998334	0.001666

2 Algorithmic Flowchart for Finite Difference Approximations

Below is the computational workflow outlining the steps used to calculate and analyze the finite difference errors and their convergence behaviors.



3 Analysis of Method Convergence

To rigorously assess the convergence behavior, calculations were systematically repeated using progressively refined step sizes: $h = 0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001$. The mean absolute error aggregated over the three spatial points was computed for each resolution and is documented in Table 2.

Table 2: Mean absolute error values recorded for progressively decreasing step sizes.

h	Mean Forward Error	Mean Backward Error	Mean Central Error
0.100	2.93760×10^{-2}	2.85907×10^{-2}	9.47919×10^{-4}
0.050	1.44600×10^{-2}	1.42636×10^{-2}	2.37069×10^{-4}
0.020	5.72810×10^{-3}	5.69667×10^{-3}	3.79349×10^{-5}
0.010	2.85464×10^{-3}	2.84678×10^{-3}	9.48388×10^{-6}
0.005	1.42496×10^{-3}	1.42299×10^{-3}	2.37098×10^{-6}
0.002	5.69415×10^{-4}	5.69100×10^{-4}	3.79357×10^{-7}
0.001	2.84613×10^{-4}	2.84534×10^{-4}	9.48393×10^{-8}

Assuming a general error model of the form $E = Ch^p$, we can linearize this relationship by applying a logarithmic transformation:

$$\log E = \log C + p \log h$$

Plotting this on a log-log scale directly reveals the convergence order p as the geometric slope of the best-fit line. Analyzing the numerical data gives approximate slopes of: $p_F = 1.006$, $p_B = 1.001$, and $p_C = 2.000$.

Therefore, we can conclude the asymptotic behaviors are:

$$E_F = \mathcal{O}(h), \quad E_B = \mathcal{O}(h), \quad E_C = \mathcal{O}(h^2)$$

This confirms that while forward and backward differences are strictly first-order accurate, the central difference scheme benefits from second-order accuracy.

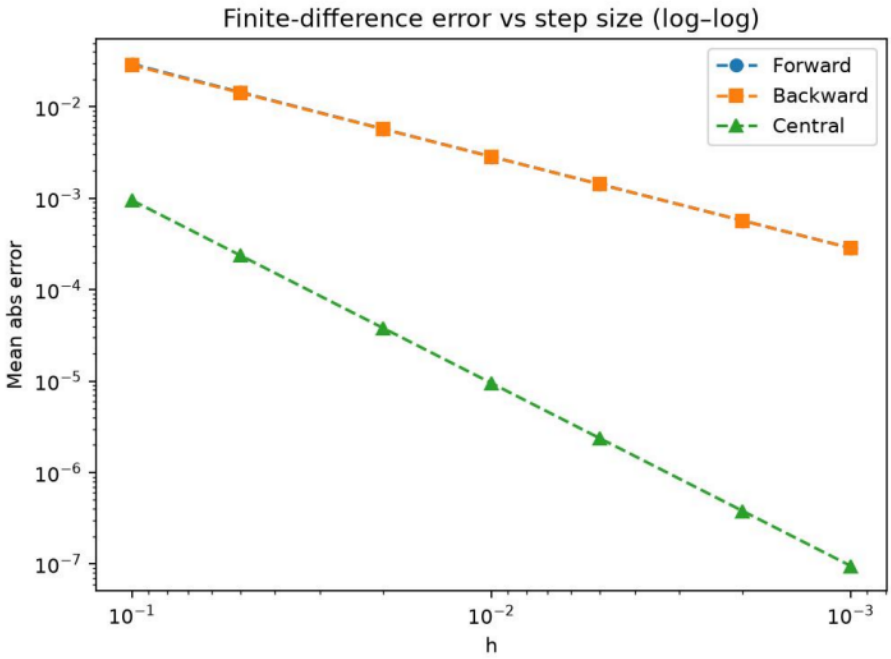


Figure 1: Log–log plot of mean absolute error versus step size for the forward, backward, and central finite-difference approximations of the derivative of $f(x) = \sin x$.

4 Algorithmic Flowchart for Root Finding

Below is the modified and colorized workflow illustrating the process of utilizing the Newton-Raphson numerical method to locate roots.

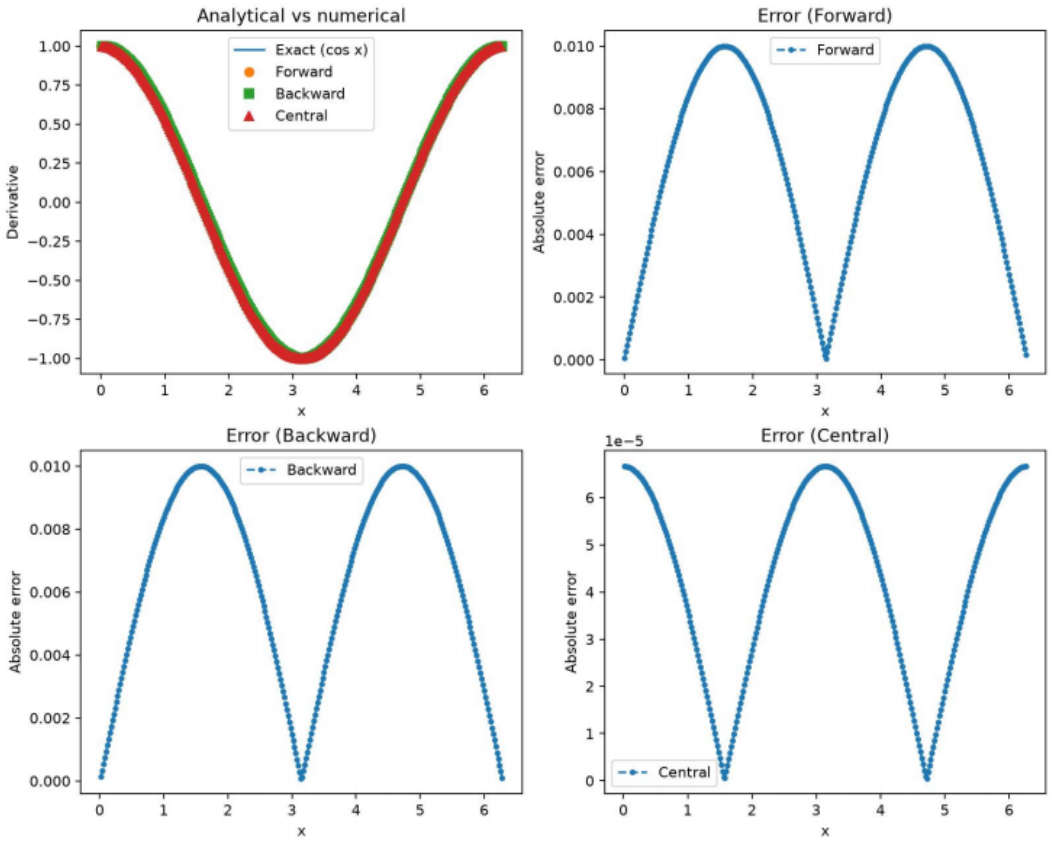


Figure 2: Comparison of analytical and numerical derivatives of $f(x) = \sin x$ using forward, backward, and central finite-difference methods, together with the corresponding absolute error distributions over the domain $0 \leq x \leq 2\pi$.

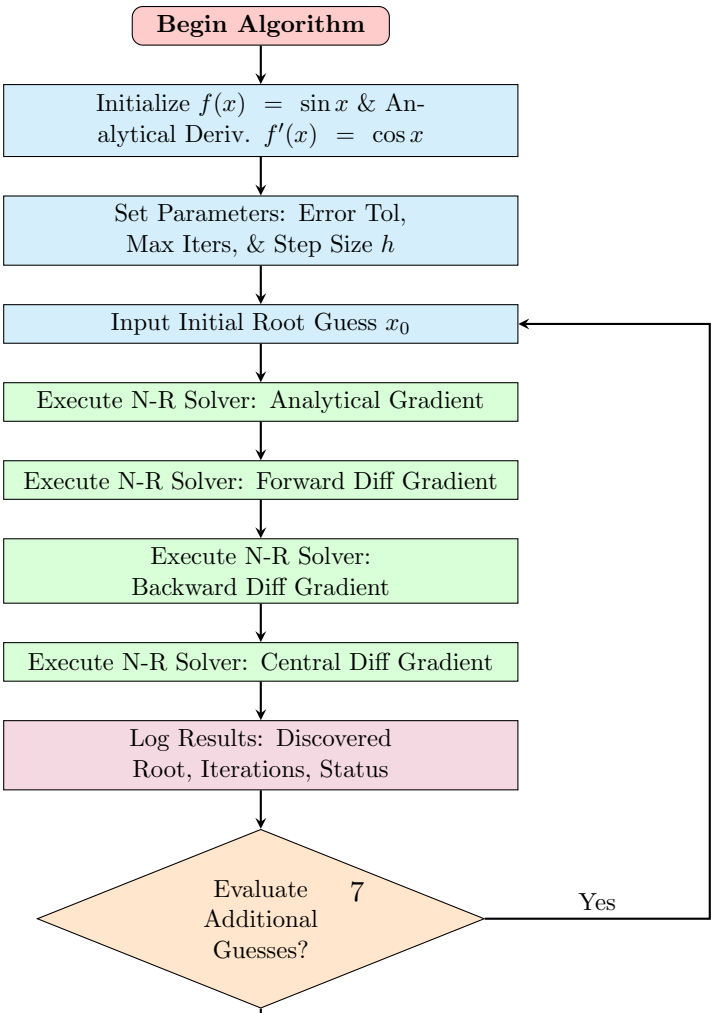


Figure 3: Enhanced algorithmic diagram mapping the execution of the Newton-Raphson procedure using both precise analytical and approximate numerical derivatives.

5 Key Conclusions

Based on the experiments conducted with the numerical differentiation of $f(x) = \sin x$, we can summarize the main takeaways as follows:

- **Methodology Comparison:** The study successfully implemented and compared three fundamental finite difference methodologies: forward, backward, and central approximations.
- **Superiority of Central Difference:** Holding the increment step size constant, the central difference method consistently registered smaller margins of error when benchmarked against forward and backward schemes.
- **Convergence Rates:** The detailed convergence analysis explicitly confirmed that both forward and backward difference algorithms exhibit first-order accuracy bounds.
- **Second-Order Accuracy:** Conversely, the central difference methodology showcased an accelerated convergence profile, acting as a second-order accurate numerical tool.
- **Error Scaling Laws:** Practically, this mathematical behavior implies that refining the step size by a factor of 10 decreases error magnitudes proportionally by 10 for first-order approximations.
- **Exponential Gain:** In stark contrast, applying that same factor-of-10 refinement to the central difference scheme mitigates the computational error by an impressive factor of one hundred.